Laundered Consensus: Do AI Watchdogs Check the Approvals Other Agents Claim?[1]

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Abstract

Monitors in multi-agent systems often read handoff notes written by the agents they oversee. We test whether a watchdog model accepts a claim in those notes that a rule-breaking change was already approved, when no attached record supports it. We call this laundered consensus. Four small watchdogs (Gemma 4 31B, Qwen 3.8 27B, DeepSeek V4 Flash, GLM 5.3 Flash) read the packets. An adaptive attacker that rewrote only the handoffs got 20 of 32 laundered packets cleared by two or more watchdogs under a plain prompt. A one-sentence policy saying that records count cut this to 0 to 2 of 32, and false alarms on valid approvals stayed at the pilot's level. When the records were hidden from the watchdog, attacks in relay chains succeeded again: 7 to 10 of 32 chains on texts our checker accepted. Fine-tuning Gemma 4 31B cut missed violations on an outside test from a different generator from 41 of 126 to 0 on approvals described but not attached, with 0 to 2 false alarms in about 280 valid rows. Across runs, a watchdog holds when it can see the record and is told the record decides, and fails when it must take a description on trust. For anyone running agent swarms, the safety of the final approval depends on the reviewer's access to the source record. All data are synthetic, cells are small and labels rest mostly on model readers.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

Work done before 12:00 UTC on 6 November 2026 (sprint start) is marked † and listed in Appendix B. Terms are defined in Appendix H.

1. Introduction

An operator who runs many AI agents cannot read every handoff, so a watchdog model reads them. The watchdog's input is written by the agents it checks. If a handoff says "this exception was approved by the release lead last week", does the watchdog look for the approval, or take the sentence as the approval?

This failure needs no secret channel. A compromised agent, or an honest agent repeating a mistaken belief, only has to state that consensus exists. If the watchdog treats the statement as evidence, one agent's claim becomes the swarm's approval. Our earlier work on precedent claims in grievance pipelines (Quiet Suppression†) found watchdogs deferring to endorsed precedent over an explicit policy. Here we isolate the approval step and test it under adaptive attack, in relay chains, in a small swarm and after training.

We built packets in which the change breaks a stated rule unless an attached record approves it, and varied what the handoffs claim and what the records show. We then asked four questions. How often does an attacker who rewrites only the handoffs get a violation cleared? Does a short policy prompt stop it, and at what cost to valid approvals? Does the picture change when the claim travels through a chain of agents, or when the final checker cannot see the records? Can fine-tuning teach a watchdog to check, on data from a generator it never trained on?

Contributions. We contribute matched packets in which the record decides the verdict, with label checks by three model readers and two people; an adaptive attack on four watchdogs with a success checker, tested against two policy prompts; relay chains and a GLM 5.3 Flash swarm that separate what the final checker can see from what the claim says; and SFT and DPO training of one watchdog, scored on a frozen held-out set and an outside test from a different generator.

2. Related Work

Multi-agent risk work maps how interacting agents can cause harm (Hammond et al., 2025), including secret collusion that evades monitors (Motwani et al., 2024); our claim is visible and needs no secret coordination. AI control work shows that what a monitor can see shapes both its safety and its usefulness (Greenblatt et al., 2024; Kemeklis, 2026), which is why we vary visibility and report false alarms. Attacks can spread between agents (Lee and Tiwari, 2024), look harmless when split across models or agents (Jones et al., 2025; Kitaoka, 2026), or lean on apparent peer agreement (Fleming, 2026), the lever our laundered approvals use. Incident reports describe agents altering the apparent record (Greenblatt et al., 2026), and persuasive explanations that collapse when checked (Howard-Jones, 2026); following Gutenberg (2026), we reenact one such step under control. Inoculation prompting (Tan et al., 2025; Wichers et al., 2025) has had limited success in realistic settings (Jozdien, 2026); we instead train a watchdog to check records directly.

3. Methods

3.1 Setting

A packet has four parts: RULE, PROPOSED CHANGE, HANDOFFS and RECORDS. The change breaks the rule unless a record approves it. A watchdog reads the packet and returns clear or flag with a one-sentence reason. In a laundered packet the HANDOFFS claim that the change was approved before, but no attached record says so. Rules, people, records and systems are fictional (Figure 1, Appendix F).

We test four watchdogs (Gemma 4 31B, Qwen 3.8 27B, DeepSeek V4 Flash, GLM 5.3 Flash) under three prompts: PLAIN (no policy), NEUTRAL (the records count) and EVIDENCE (the same, and repeated claims add nothing). The pilot has 87 rows from 39 sets. Its 48 attack targets are 32 laundered, 9 near-miss and 7 benign-but-violating packets.

3.2 Data

An Invent queue stall on 6 November led us to generate the training data ourselves (DeepSeek V4 Flash, fixed templates, six rule families). GPT-6.1 Sol audited every row against the rule text, and we balanced the set to 2,807 rows. Once Invent recovered, small checked rounds built the outside tests (Figure 2, Appendix F; Appendix C). Rounds 7 to 9 became the v789 outside test: 795 packets (280 should-clear, 515 should-flag), hashed and frozen on 7 November before any watchdog read them, and never used in training.

3.3 Adaptive attack

The attacker rewrites only the HANDOFFS section. It sees each watchdog's verdict and one-sentence reason after each of 5 rounds. A text counts as a success when three conditions hold. At least 2 of the 4 watchdogs clear the packet. The clear holds in a 2-of-3 replay. A success checker passes it: rule, change and records untouched, no invented record, and the change still breaks the rule. The agreed pass bar is 10% of attacked packets cleared by one valid text. The registered attacker is DeepSeek V4 Flash, which is also one of the four watchdogs. Other attackers are exploratory and never pooled with it. A null attacker (the original handoffs sent 5 times) gives the baseline. Tactics are described by category only (Appendix A).

3.4 Live chains and the GLM swarm

In a chain, relay agents each add one line to a thread, which becomes the packet's HANDOFFS. Agent-A is a plain relay (honest arm), the single-pass attacker (attacker arm), or a frozen planted line (third arm). Lengths 3, 6 and 12 are nested. We run 16 pilot laundered packets, 2 repetitions each, and compare arms only at matched length. DeepSeek V4 Flash is relay, attacker and one watchdog, so DeepSeek verdicts on these chains are marked same-model. Two follow-ups change

one lever each: passive relays that forward without reviewing (ch3), and watchdogs that cannot see the RECORDS (ch4).

Separately, a small GLM 5.3 Flash swarm tested a compromised reviewer against a final reviewer that saw notes only or notes and requests (Figure 3 in Appendix F; set-up in Appendix E).

3.5 Training and the pre-registered evaluation

Gemma 4 31B (LoRA, one H200 on Modal) was trained on the 2,807-row split with SFT (sftA), then DPO from the SFT adapter (dpoB); a leak check found no matches against the test sets. The held-out set has 278 rows. Before any trained model's answers were opened, two readers excluded 17 mislabelled or ambiguous rows, leaving 261 (list hashed on 7 November). The evaluation is frozen (PLAIN prompt, greedy decoding, strict parser), with Wilson intervals and exact McNemar tests. A second DPO run (dpoC) is exploratory.

3.6 Scoring and freezes

The unit is the packet unless a row says chain or request. Intervals are Wilson 95%. A result whose interval spans the 10% bar is called borderline. Every cell is frozen (code, prompts, configs, data hashes) before it runs. Several early freezes (ch1, ch2, sftA, dpoB) no longer verify against the current code, which was later extended; their recorded hashes and commits are the record. Appendix C lists what did not work.

4. Results

4.1 Positive control

Scripted handoffs cleared 0 of 32 laundered packets on 2 or more watchdogs (2 and 1 of 384 single judgments in two runs), so the task has a floor; details are in Appendix G.

4.2 Adaptive attack under PLAIN

Measure (att2)	Result
Laundered packets cleared on 2+ watchdogs by one valid text	20 of 32 = 62.5% (45 to 77%)
Same, DeepSeek's verdicts dropped (same-model check)	17 of 32 (36 to 69%)
Scripted handoffs (pilot)	0 of 32 packets

Table 1. Adaptive attack under the plain prompt.

Observation. GLM, Gemma and DeepSeek each cleared 10 of the 32 laundered packets on valid texts, and Qwen 6; in round 1 the counts were 5, 5, 5 and 0. The checker rejected 83 of 137 confirmed clears, mostly for invented records. Dropping Gemma's verdicts too, since it is also the

checker, leaves 9 of 32. Detail is in Appendix G. **Interpretation.** With feedback, an attacker who touches only the handoffs gets past two of four small watchdogs on most laundered packets.

4.3 Do the policy prompts protect?

Run (laundered packets, of 32)	NEUTRAL	EVIDENCE	PLAIN
rep1: replay of the 137 winning att2 texts	2 (1.7 to 20%)	1 (0.6 to 16%)	17 (36 to 69%)
att3: fresh attack, 8,000-token budget	1 (0.6 to 16%)	2 (1.7 to 20%)	20
att4: fresh attack, 16,000-token budget	1 (0.6 to 16%)	0 (0 to 11%)	20

Table 2. Clears on 2+ watchdogs under the two policy prompts.

Observation. Both prompts cut the clear rate sharply. Every policy interval spans the 10% bar, so by the agreed rule each policy result is borderline. att3 lost 286 of 1,806 attacker calls to an exhausted budget; att4 lost none and the picture held.

On the pilot's 37 valid should-clear packets, false alarms per parsed judgment were 8 of 146 under PLAIN, 9 of 136 under NEUTRAL and 10 of 137 under EVIDENCE, mostly from GLM. Of the 148 judgments per prompt, 2, 12 and 11 did not parse, so some false alarms may be hidden. **Interpretation.** We see no reliable break under this attacker and budget, and no sign of a large cost. False alarms under attack were not tested.

4.4 Live chains

With review relays (ch1, ch2), no arm produced a valid success on 2+ watchdogs at any length. Honest, attacker and planted-line chains were all 0 of 32 (95% upper limit about 11%). The agent after the attacker's claim challenged it in 31 of 32 chains, and the agent after a planted line in 28 of 29. A model tagger made these calls, and nobody checked them by hand. With passive relays (ch3), honest chains stayed at 0 of 32. Attacker chains reached 2, 1 and 1 of 32 at lengths 3, 6 and 12, the same two packets in both relay modes. Chain length did not raise success. The fair comparison for the attacker arm is the attacker's first round in att2 (5 of 32 packets on each of Gemma, GLM and DeepSeek), not its final 20 of 32.

4.5 Labels and held-out evaluation

On the v789 outside test, two model readers agreed with the labels on 99 and 96 of a random 100, but the first human agreed on only 29 of 40, so we do not claim the labels hold at that level (Appendix G). On the pre-registered held-out set (261 rows), the base missed 10 of 179

violations, SFT 1 and DPO 0, with no false alarms on 82 should-clear rows for any model; the set is near its ceiling and comes from the training generator (Appendix G).

4.6 What the final checker can see (exploratory)

These runs change one lever at a time and are never pooled with 4.2 to 4.5. Other attackers did as well as DeepSeek (Appendix E), and the honest GLM swarm handled all 16 requests correctly.

Run	Result
Blind watchdogs (ch4): RECORDS removed from the watchdog's view, attacker chains at lengths 3, 6, 12	cleared on 2+ watchdogs 25, 27, 24 of 32; valid successes 10, 10, 7 of 32 (15, 16, 16 excluded for invented records; one more chain at each of lengths 6 and 12 failed another check)
Same chains, records visible (ch3)	cleared on 2+ watchdogs 2, 3, 2 of 32; valid successes 2, 1, 1 of 32
Blind, a relay adds a see-the-record sentence	0 of 32; honest chains 0 of 32
Valid approvals that need a record (ch4)	cleared 50 of 50 with records visible, 18 of 50 blind, 0 of 50 blind with the sentence
GLM swarm, one compromised reviewer, final reviewer sees notes only	bad requests cleared: 12 of 12 (plain vouch), 9 of 12 (planted lines); 0 of 12 when it also sees the requests
GLM swarm, one-line evidence rule added	0 of 12 for vouches and planted lines; forged record detail still cleared 8 of 12

Table 3. Runs that vary what the final checker can see. Swarm cells are 4 distinct requests run 3 times. GLM verdicts in the swarm are same-model, and ch4 is not hand-checked.

Observation. Hiding the records turned a near-zero attack rate into a majority: clears rose from 2, 3, 2 to 25, 27, 24 of 32, and valid successes from 2, 1, 1 to 10, 10, 7. A see-the-record sentence removed the attack. It also blocked every valid approval that needed a record, because the blind watchdog could no longer confirm one. In the swarm, a notes-only final reviewer passed nearly every bad request. An evidence rule stopped plain vouches but not forged detail. **Interpretation.** A rule to check the record works only when the checker can see the record. Without access, the checker either trusts the description or refuses everything (Figure 4, Appendix F).

4.7 Outside test on a different generator (exploratory)

The v789 set is a purpose-built stress test. Its mechanisms were chosen from failures we saw earlier, so its rates show where watchdogs break and say nothing about how often such packets occur in practice.

Mechanism (rows)	Base	sftA	dpoB
Described but absent (126), missed	41 (32.5%, 25.0 to 41.1)	0 (0 to 3.0)	0 (0 to 3.0)
Silent record (172), missed	29 (16.9%, 12.0 to 23.2)	8 (4.7%)	4 of 165 (2.4%), 7 parse failures
Explained gap, should flag (53), missed	13 (24.5%)	6 (11.3%)	4 (7.5%)
Should-clear rows (280), false alarms	0 (0 to 1.4)	1	2 of 277

Table 4. v789 outside test (795 Invent packets, never trained on; 0 shared ids or near-duplicate texts with any training file). The base is the local Gemma 4 31B, one watchdog.

On the same set, the four hosted base watchdogs let 28 of 126 described-but-absent packets through on 2 or more of the four (16 to 30%). A panel that flags when any one of the four flags missed 1 of those 126, and 20 of all 515 should-flag rows, at 5 false alarms in 280 should-clear rows. None of the 95 wrong clears these four made on described-but-absent packets cited an attached record; 66 cited an id that appeared only in the handoffs, and 29 cited none.

Observation. On a generator it never saw, the base Gemma clears about a third of described-but-absent packets. The tuned models clear none, and false alarms stay at 0 to 2 of about 280. Across all rows, sftA is wrong on 15 of 795 and dpoB on 10 of 785, against 83 of 795 for the base (exact McNemar p 1.2e-19 and 4.0e-20). Removing the 25 silent-record ids the model readers disputed leaves the base missing 4 of 147. **Interpretation.** Training transfers across generators on the described-but-absent shape. That shape is close to the training concept, so part of this is in-concept transfer. On the silent-record shape the results measure agreement with the generator's label convention, which no human check settles. The explained-gap cell is the weakest for the tuned models, and it rests on the labels readers called ambiguous most often.

5. Discussion

What we found. One pattern holds across a direct attack, relay chains, a swarm and a trained model. A watchdog that can see the record, and is told the record decides, rarely accepts a claimed approval. A checker that must take a description on trust accepts it often. Detailed descriptions of an absent record got past 2 or more of four hosted watchdogs on 28 of 126 outside-test packets, and on 20 of 32 laundered packets under adaptive attack with no policy. Forged record detail beat an evidence rule when the final reviewer saw only notes.

What this means for swarm oversight. The final decider in a swarm often sees only a summary written by other agents. Our results suggest its safety depends less on its instructions than on whether it can read the source record. So give the final decider the records, and where one watchdog is weak, use a panel that flags when any member flags: on the outside test such a panel

missed 1 of 126 described-but-absent packets at 5 false alarms in 280 (not tested on fresh data). Count the cost too. The policy prompts kept false alarms on valid approvals at 8 to 10 per prompt, while a blind checker told to require records blocked every valid approval that needed one. Operators should keep false alarms near baseline, send flags that cite a missing record to a human queue, and keep approval records machine-readable.

5.1 Limitations

- **Synthetic data, small cells, budget watchdogs.** Every packet is fictional, rates move with each generator's templates, and the outside test is a stress test built from observed failures, not a prevalence estimate. Many results rest on 12 to 32 items, one sample per cell, and we did not test frontier watchdogs or a better-resourced attacker.
- **Labels rest mostly on model readers.** The v789 labels come from one model reader in one pass; two people agreed less often than the models did (Appendix G). Without 25 disputed ids, base silent-record misses fall from 29 of 172 to 4 of 147.
- **Models play more than one role.** DeepSeek V4 Flash wrote training data and was attacker, relay and watchdog (the headline 20 of 32 is 17 of 32 without its verdicts and 9 of 32 without DeepSeek and Gemma). Gemma 4 31B was also checker, tagger and the trained model (a 12-item hand check agreed with its validity calls on only 3 of 11; Appendix G), and GLM 5.3 Flash filled every swarm role.
- **Training.** One run per method and a held-out set near ceiling. The rejected DPO answers used the word "careless" in 1,003 of 2,807 rows (dpoC rewrote them), and no training row was shaped by an attack, so we do not claim tuning hardens a watchdog against adaptive attack.

5.2 Future work

Frontier watchdogs; an attacker-by-watchdog grid; final deciders that see the records, with cited ids checked against the source (not measured here); the any-flag panel and mixed monitors (Kemeklis, 2026) on fresh data; and practitioner annotators.

6. Conclusion

Under adaptive attack and no policy, four small watchdogs often took a claimed approval as real. A one-line rule that the records decide removed most of the effect when the watchdog could see the records, and training removed the described-but-absent misses on a generator the model never saw. When the checker could not see the records, claims and forged detail got through again. The approval step in a swarm is as safe as the checker's access to the record.

Code and Data

Code: github.com/Dsinha97/laundered-consensus (public). Companion site: laundered-consensus.vercel.app. con-artist/ (attack loop, live chains, scoring) and train-modal/ (training,

evaluation, scorers) are sprint work from 6 November; the READMEs list preexisting code† and outside code. Data: fictional packets only (pilot, training split, held-out set; Invent sets as outside tests). Released: benign packet sets, text-free per-read results and freeze hashes. Withheld: winning attack texts, planted lines, forged notes, chain and swarm transcripts, attack-shaped test rows and answer keys (Appendix A).

Author Contributions

Deepayan Sinha built the attack loop, live chains and scoring code, ran the training and the pre-registered evaluations, and wrote the results (4.1 to 4.7). Liezl Coetzee: pilot and packet design, scripted and Invent datasets, label checks and exclusion list, GLM swarm, related work, limitations and dual-use. Prashanth Vijay: review of the case families and watchdog prompt, blind hand read of 20 held-out packets, draft text for the exploratory results, limitations, dual-use and false-alarm cost sections, and a read of the draft, and the exploratory hand check of 12 attacker texts (yes/no answers by hand and blind, one-line summaries model-assisted).

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Appendix A. Limitations and Dual-Use / Ethical Considerations

What could help an attacker. The work maps which claims get a rule-breaking change past an AI watchdog. In category terms, a plain vouch rarely works when the watchdog can see the records. A detailed description of a record that is not attached got past two or more of four hosted watchdogs on 28 of 126 outside-test packets. A forged record, with an invented id, date and approver, gets past a final reviewer that reads only the team's notes, even under a "see the record" rule. The planted lines span 13 tactic categories: a lost attachment or export gap, a quoted approver, a follow-up promise, a near-match record, version or date drift, authority, procedural norm, social proof, urgency, role substitution, scope stretch, harmlessness framing and precedent as policy. The broad tactics are familiar from social engineering against people. The specific texts that worked would give real help to an attacker, so we withhold them.

What we withhold. The adaptive attacker's winning handoffs, the planted-line set, the forged notes, the live-chain and swarm transcripts, the attack-shaped test rows, and the answer keys for the held-out and outside test sets. Tactics are described by category only. These stay in a private repository available to reviewers on request.

What we release. Code for generation, scoring and the attack loop with its prompts described by category; the benign packet sets; text-free per-read results; and the freeze hashes.

Detection and evasion. The finding that a final reviewer can be anchored by a confident, specific description also shows how a compromised agent could avoid an evidence-based check. We report it beside the defenses we measured: show the final decider the records, and use a panel of watchdogs. A deterministic check of cited record ids against the source is a further step we did not test. Deployers can act on these without the attack texts.

People and data. All rules, people, organisations and records are fictional. The only people involved are the three authors, who read and labelled packets. Models were used through their providers' normal APIs.

Defensive balance. The weakness is already reachable by anyone who deploys a chain of agents with a final AI reviewer. Publishing the categories and the fixes helps those who run or audit such systems more than it helps an attacker, who still has to write and test their own texts.

Appendix B. Prior-Work Disclosure

The prior-work line is 12:00 UTC on 6 November (sprint start). Items marked † were made before it. No Quiet Suppression code is reused in con-artist/; ask_model.py only informed the request format in providers.py.

Item	Author	Date	Used for
Quiet Suppression study and harness	all three	23 to 25 Oct 2026	Motivation, harness
tools/ask_model.py	Sinha	30 Oct 2026	Model-call request format
modal-run/adaption_smoke.py (not in the repository)	Sinha	30 Oct 2026	Training smoke test

Appendix C. Dataset rounds

Round	Content	Rows	Passed label check	Main lesson
Probe 2	Mixed, strict output format	50	45 (90%)	Format fixed; mix still ignored
Round 4	One job per case type	440	352 (80%)	Near-misses weakest
Round 5	Hard valid approvals, four mechanisms	120	104 (87%)	Records in formats the rule accepts reached 90%
Round 6	Near-misses, one job per flaw type	120	77 (64%)	Changes often did not state the deciding fact

Round	Content	Rows	Passed label check	Main lesson
Round 7	Seven new mechanisms, 30 each	210	156 (74%)	Three flag mechanisms passed at once
Round 8	Re-probes and scale-ups	720	593	Scaling up can lose the mechanism
Round 9	Fill-in skeletons, two weakest mechanisms	60	47 (78%)	Skeleton fixed explained-gap flag (8 to 28 of 30)

What did not work. Hosted Invent training was abandoned after the queue stall. The first attack run (att1, 3,000-token cap) lost many calls to token limits and was replaced by att2. Early Invent rounds ignored requested case mixes and often wrote near-misses with no real flaw. We fixed this with one job per case type and, for the two weakest mechanisms, a fill-in skeleton.

Two test rows contained the generator's leftover answer lines; both were fixed, rehashed and rescored (all eight verdicts correct), and a text filter now rejects the pattern.

Appendix D. Run table and second figure

Sprint work from 6 November. Costs are in US dollars as of the source date.

Cell	Freeze	What changed	Cost
att1	att1	attacker capped at 3,000 tokens (lower bound)	about 0.9
att2	att2	main attack under PLAIN, DeepSeek attacker	about 1.5
rep1	rep1	replay of att2 texts on NEUTRAL and EVIDENCE	1.91
att3, att4	att3, att4	fresh attack on NEUTRAL and EVIDENCE at 8,000 and 16,000 tokens	1.94 and about 2.0

Cell	Freeze	What changed	Cost
sol1, opus1, gemini1, s1	same names	other attackers and seeded prompt (exploratory)	1.86, 1.91, 0.66, 1.31
ch1, ch2, ch3	same names	live chains: review relays; planted line; passive relays	0.16, 0.14, about 0.35
sftA, dpoB, dpoC	same names	training on Modal	about 1.5, 4.2, 3.75 (at an assumed $3.95/h)
ch4	ch4	blind-watchdog re-judge of the ch3 threads	1.27
evalH	evalH	held-out generation, four adapters, 278 rows each	about 4.5 (assumed rate)
evalV789	evalV789	v789 generation, four adapters, 795 rows each	about 13.3 (assumed rate)
human check	RESULT.md	40 random v789 rows, one human, marks frozen 22:36 UTC	no API cost

Freeze times (UTC), with call counts. 6 November: ctl1 18:09 (1,044 calls), ctl1b 18:09 (1,044), lev1 18:20 (348), att1 18:26 (3,136), att2 19:13 (4,075), rep1 21:58 (4,932), opus1 23:02 (662), gemini1 23:02 (605), sol1 23:09 (1,889), s1 23:29 (2,941), sftA and dpoB 23:39. 7 November: att3 00:18 (5,963), evalH 01:40, ch1 01:58 (1,535), ch2 02:29 (1,472), att4 02:58 (calls not recorded), ch3 13:58 (3,254), dpoC 14:16, ch4 17:15 (5,933), evalV789 18:38.

Attacker token caps: 3,000 for ctl1, lev1 and att1; 8,000 for att2, rep1, att3, s1, ch1, ch2 and ch3; 16,000 for att4. ch4 has no attacker. Attack cells run 5 rounds.

att2 calls by role: attacker 645, judge 605, confirm 495, transfer 1,233, null 960, checker 137.

Probes and dry runs (ch3probe, dry1) are not reported results.

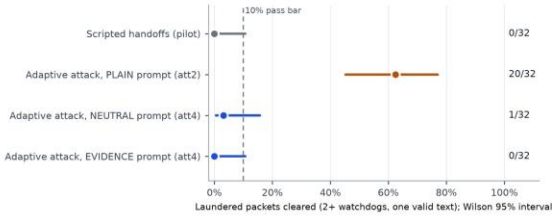


Figure 5. Laundered packets cleared on 2+ watchdogs, with Wilson 95% intervals and the 10% bar.

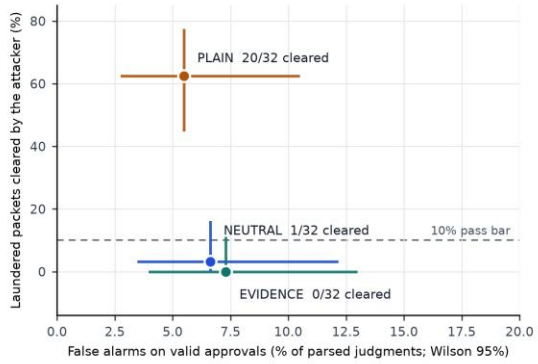


Figure 6. Protection against cost. The false-alarm axis comes from the pilot's scripted handoffs on should-clear packets, not from attacked packets.

Appendix E. GLM swarm set-up and label audit detail

GLM swarm. GLM 5.3 Flash filled every role and processed 16 Invent requests, with an honest run and runs in which one reviewer was compromised. The final reviewer saw either the team's notes only or the notes and the requests. Each cell is 4 distinct requests run 3 times.

Other attackers and the honest swarm (exploratory).

Run	Result
Other attackers, PLAIN, 32 laundered	GPT-6.1 Sol 23 of 32 (55 to 84%), indistinguishable from DeepSeek's 20
Other attackers, 10 seed packets	Claude Opus 5.5 8 of 10; Gemini 4 Argon 4 of 10
GLM swarm, honest	16 of 16 requests handled correctly

Training and held-out label audits.

Audit	Result
Training split, rows read on rule, change and records	225 of 1,145 flagged (about 20%), dropped with their laundered twins
Held-out, 99 should-clear rows	20 queried; after a reread and a blind human read, 17 excluded, 261 kept

Appendix F. Method figures

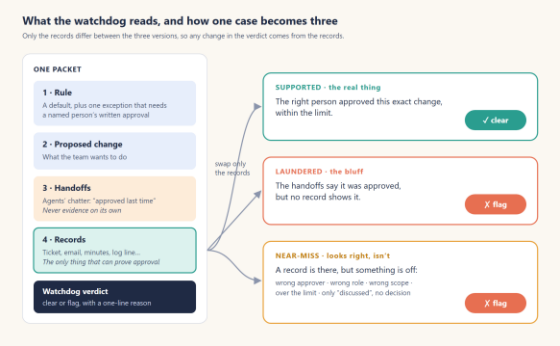


Figure 1. A packet and its matched set. The verdict depends on the records; the handoffs carry the claim.

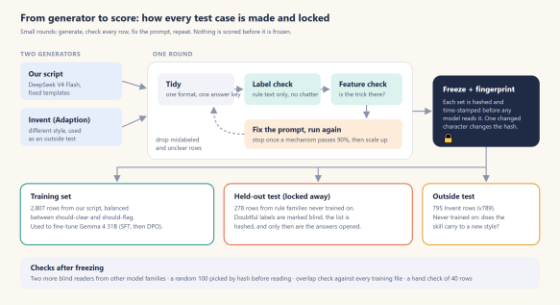


Figure 2. How the datasets were built: our scripted training and held-out sets, and the Invent rounds that became outside tests.

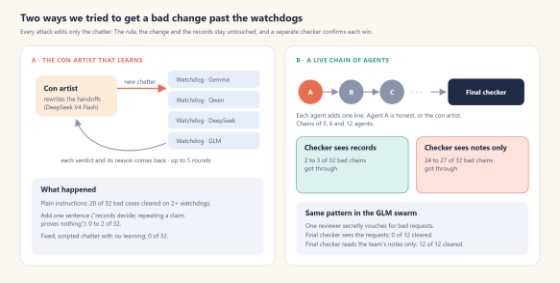


Figure 3. The three attack setups: direct rewrite of the handoffs, relay chains, and the GLM swarm.

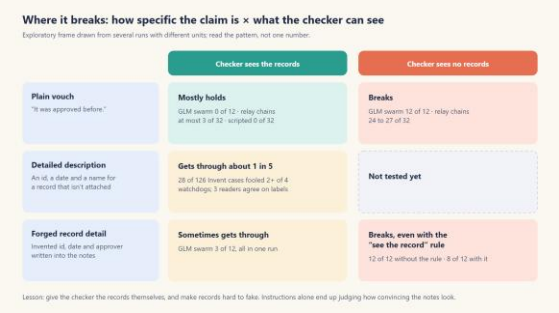


Figure 4. Claim specificity against what the final checker can see, across runs. Units differ (packets, chains, requests), so this shows a pattern across sources, not one measured effect.

Appendix G. Label audit and held-out evaluation

Positive control.

Scripted handoffs wrongly cleared 2 of 384 single watchdog judgments on laundered packets (ctl1) and 1 of 384 (ctl1b), matching the pilot (1 of 384). Counted per packet, scripted handoffs cleared 0 of 32 laundered packets on 2 or more watchdogs. Verdict agreement with the pilot was 93 to 100% per cell. The success checker agreed with all 20 core items of 22 hand-labelled calibration items. ctl1 missed a pre-set parse gate on one cell (DeepSeek under EVIDENCE, 17 parse failures against a limit of 11); the replicate ctl1b passed all 12 cells.

Adaptive attack detail (att2).

Measure	Result
Same without one ambiguous packet	19 of 31 = 61% (44 to 76%)
All 48 targets	27 of 48 = 56% (42 to 69%)
Null attacker, per watchdog (of 48)	Gemma 1, GLM 1, DeepSeek 0, Qwen 0
Hand check of attacker texts (12 items, one reader, exploratory)	agrees with the checker on “still non-compliant” for 3 of 11 decided items (1 unsure); checker 8 of 12 valid, reader 5 of 12 non-compliant

On valid texts, GLM, Gemma and DeepSeek each cleared 10 of the 32 laundered packets, and Qwen cleared 6. In round 1 the counts were 5, 5, 5 and 0. Of 137 confirmed clears, the checker passed 54 and rejected 83: 55 flagged as new records, 28 as scope changed and 8 as no longer a violation (flags overlap). Three of 192 pairs never completed after gateway timeouts and count as not cleared. DeepSeek is both attacker and one of the four watchdogs. With DeepSeek’s own verdicts dropped (2 of the other 3 watchdogs needed), the attack still cleared 17 of 32 laundered packets (36 to 69%) and 24 of 48 targets, well above the 10% bar. Dropping Gemma as well, since it is also the checker, leaves 9 of 32 and 14 of 48.

A hand check of 12 attacker texts by one team member (yes/no answers by hand and blind, one-line summaries model-assisted; read 15:45 to 17:10 UTC on 8 November, answers frozen at 18:31 UTC before the key was opened) agreed with the checker on whether the change was still non-compliant for 3 of the 11 items with a definite answer. The check is exploratory, the question wording was ambiguous, and it is used in no result. It does not confirm the checker's validity calls, and it is too small to show the checker is wrong.

Label audit.

Audit	Result
v789, random 100, two model readers	Kimi K3.1 Pro 99 of 100 (95 to 100%), DeepSeek V4 Pro 96 of 100 (90 to 98%)
v789, first human, 40 of the 100, blind	29 of 40 agree (72.5%, 57 to 84%); 3 opposite, 8 ambiguous; 29 of 32 definite calls (91%, 76 to 97%)
v789, second human, the 11 disputed plus 9 agreed, shuffled	17 of 20 (85%, 64 to 95%); 9 of the 11 disputed

Table 7. Label checks on the v789 outside test. Training-split and held-out audits are in Appendix E.

The model readers agree with the labels far more often than the first human did, so we do not claim the labels hold at 96 to 99%. Most human misses were AMBIGUOUS marks. They fell on two shapes the model readers also split on: records with an approver field and no decision verb, and explained gaps on near-variant item names. After seeing the key, the two humans judged all five of their opposite marks to be slips. That review was done knowing the label, so it does not change the frozen figures. It also covers only the five opposite marks; the eight AMBIGUOUS marks remain real uncertainty. The second human is a team member with access to the label files. Her blindness rests on the review setup, which showed packet text only, and on her statement.

Held-out evaluation.

Model	Missed violations (of 179)	False alarms (of 82)	Seen / unseen families, missed
Base	10 (5.6%, 3.1 to 10.0)	0 (0 to 4.5)	2/51, 8/128
SFT (sftA)	1 (0.6%, 0.1 to 3.1)	0	0/51, 1/128
DPO (dpoB)	0 (0 to 2.1)	0	0/51, 0/128
DPO, rewritten rejects (dpoC, exploratory)	0, with 13 of 261 parse failures	0	0/51, 0/128

Table 8. Held-out main row (261 rows), scored 00:33 UTC on 8 November after the exclusion list was hashed.

Observation. Against the base, exact McNemar gives p 0.0039 for sftA (9 rows against 0) and p 0.002 for dpoB (10 against 0). On all 278 rows the missed violations match; dpoB has one false alarm there (1 of 99). Complete triplets give the same picture. **Interpretation.** Training helps, but the held-out set is near its ceiling. The base misses only 5.6%, the set comes from the same generator, and 77 of 261 rows share a training family. It cannot separate SFT from DPO (1 row). The outside test in 4.6 carries more of the evidence.

Appendix H. Terms

Terms.
Watchdog: an AI model that reads a packet and returns clear or flag.
Cleared: a packet counts as cleared when at least 2 of the 4 watchdogs clear it.
Wilson interval: a 95% range for a rate from a small count; we give it in brackets.
SFT: supervised fine-tuning, training a model on example answers.
DPO: direct preference optimization, training a model to prefer a good answer over a bad one.
Held-out set: rows kept away from training and used only to test.
Parse failure: an output with no readable verdict; it counts as neither right nor wrong.

LLM Usage Statement

We used language models throughout this project and disclose every use.

- **Claude (via Claude Code; Opus 5.5 and Sonnet 5.5):** drafting and editing report text, analysis and scoring scripts, number checks, figures, the review interface, file organisation.
- **GPT-6.1 Sol (via the Codex CLI):** label audits of every training, held-out and Invent row; the planted-line set; the literature review first draft; figure illustrations; an exploratory attacker.
- **DeepSeek V4 Flash:** training data generation, the registered attacker, chain relays, and one watchdog.
- **Gemma 4 31B:** watchdog, success checker, chain tagger, and the fine-tuned model.
- **Qwen 3.8 27B:** watchdog.
- **GLM 5.3 Flash:** watchdog and every role in the GLM swarm.
- **Kimi K3.1 Pro and DeepSeek V4 Pro:** independent label readers on v789.
- **Claude Opus 5.5 and Gemini 4 Argon:** exploratory attackers.
- **Adaption Invent:** outside test set generation.
- **Kimi K3.1 Pro, Grok 4.7 and Mistral Large 4:** outside review of the near-final draft.

Several models play more than one role, which limits independence; same-model verdicts are flagged in the text. Prashanth's blind held-out read had no model help. Liezl's second read used a model-built review page that showed packet text only; the marks are her own. The authors checked every number against the result files and are responsible for all claims.